WATS
DESIGN DOCUMENT

DESIGN AND TECHNICAL GUIDELINES
FOR FRONT END DEVELOPMENT OF
THE WATS PRODUCT

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CHANGE LOG

DATE	AUTHOR	CHANGES
16.04.2020	AK	Initial release
26.05.2020	AK	Added guidelines for bold text.
03.12.2021	AK	Added default columns chapter
11.01.2021	AK & TAL	Revised default column names
24.08.2022	AK	Added UTC last seen date/time to default columns

GLOSSARY

TERM	EXPLANATION
CSS	Cascading Style Sheet: language with native browser support for designing HTML pages.
Angular	Javascript-based framework for developing a component-based application front-end.
Entity	A distinct set of data. Examples includes data about a user or data about a process.
HTML	Hypertext markup language: language with native browser support for designing HTML pages.
Module	A part of a WATS module, for instance Users, Processes or Manual inspection.
WATS application	One of the main domains in WATS. Internally, the applications are: Reporting, Operator interface and Control panel. In addition WATS is comprised of the following additional supporting modules: Start page, My settings and Account.
WATS	Product name. Originally an acronym meaning Wide Area Traceability System.
WLA	WATS Layout Action. A button or other element present in the WATS layout, which defines a global action for the layout.

SCOPE

The WATS design document presents guidelines for how to design the front-end web application for WATS. This includes all work done with the Angular framework.

The document also includes a basic overview of which default components should be utilized to obtain the desired design.

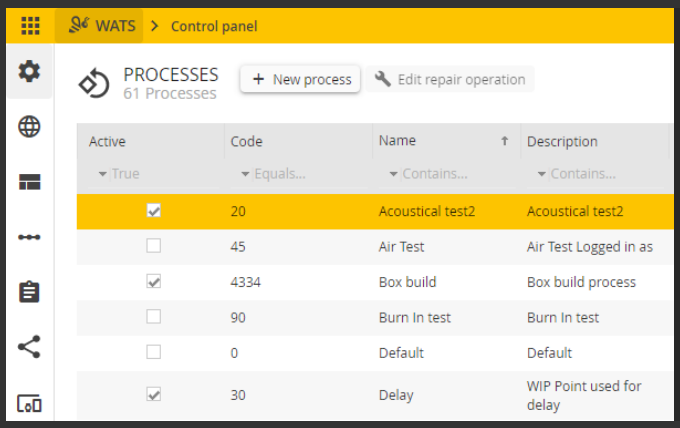
A design guideline should be considered a design rule except in the following cases:

- The product manager or team lead overrides the guideline. If the override is not considered an isolated event, the design document should be revised.
- Usability dictates that the design guideline cannot be followed. Example: Workflow execution module, designed to be displayed on screens on factory floor, requires larger text. In this case, only break the guidelines that are strictly required to maintain usability at a high level. Discuss changes with team lead before implementing.

The design guidelines in this document do not govern the WATS mobile app or the WATS client. However, it is recommended that the designs of all WATS applications do have a correlation.

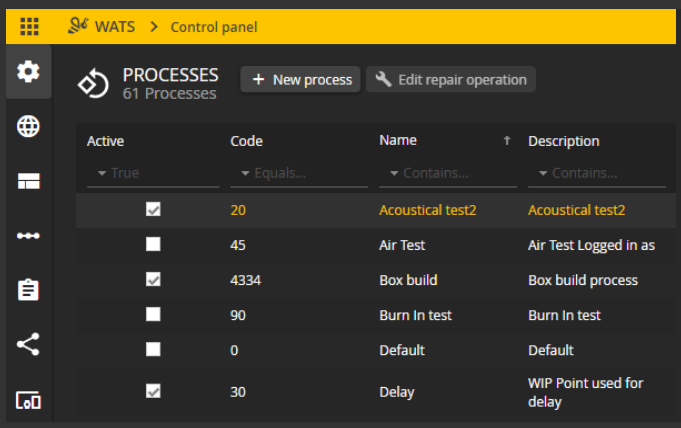
COLOR THEMES

WATS implement a total of 3 different themes available on a per user basis. Each theme is intended to improve the usability of the application. The following describes an overview of each theme:



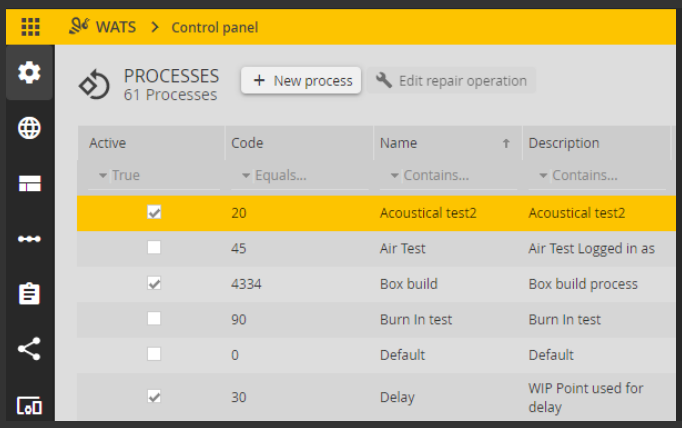
LIGHT THEME

The light theme provides dark text and icons on a white background, with some light grey design elements. This theme provides a simple look with a minimum of darker tones.



DARK THEME

The dark theme provides light text and icons on a dark grey background. This theme is suited for light sensitive people, or for those who simply prefer a darker look to the application.



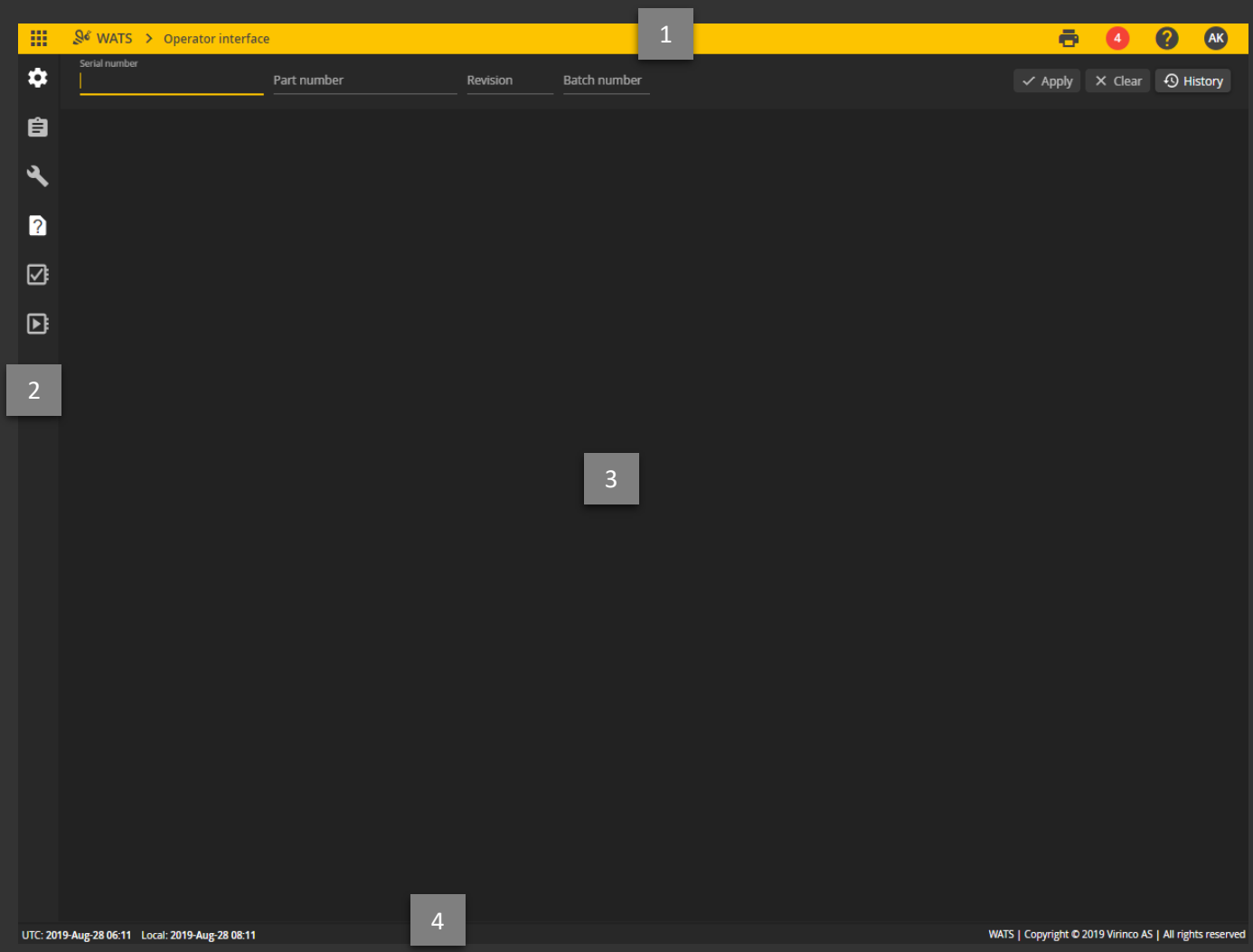
CONTRAST THEME

The contrast theme applies different styling on menu and footer than on the content. The menu and footer are borrowed from the dark theme, with light text on dark grey background. The content area uses dark text and icons on a light grey background. This theme is suited for those who feel the light theme is too bright, but do not want a completely dark theme.

MASTER GRID

The master grid is a set of CSS rules utilized by most modules in WATS. The grid defines the following major components:

#	COMPONENT	PURPOSE
1	Top bar	Quick navigation between WATS modules, user help, options and actions.
2	Menu	Navigation between modules in a WATS application.
3	Content area	Module content.
4	Footer	Clock and copyright information.



WATS
LAYOUT

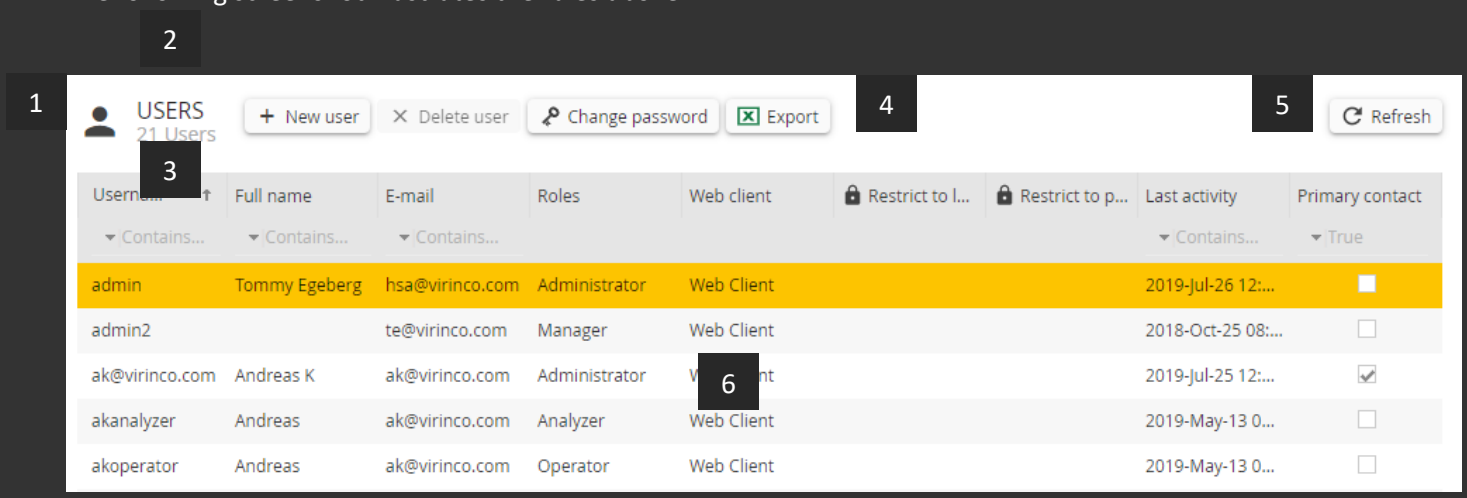
The WATS Layout is a set of Angular components developed internally by Virinco, specially for WATS. They provide both a default way of designing modules in WATS, in addition to a lot of out-of-the-box functionality. The following chapter describes the use cases for each component.

WATS LAYOUT COMPONENT (WL)

The WATS layout component is the default wrapper for modules in WATS. It provides a default design to the modules, in addition to bringing built-in functionality. It will typically occupy the whole area the Master grid content area.

#	FEATURE	PURPOSE
1	Page icon	Visual identifier for the module.
2	Page header	Display name of the module.
3	Page sub header	A field for a small peace of information, suplementing the Page header.
4	WATS Layout actions (WLA)	Actions (like buttons or selects) which is relevant for the whole module.
5	Default layout actions	Actions avalable across all layouts.
6	Content area	Where the content of the module is rendered.

The following screenshot illustrates the rules above:



WATS LAYOUT HEADER (WLH) & WATS SIMPLE LAYOUT (WSL)

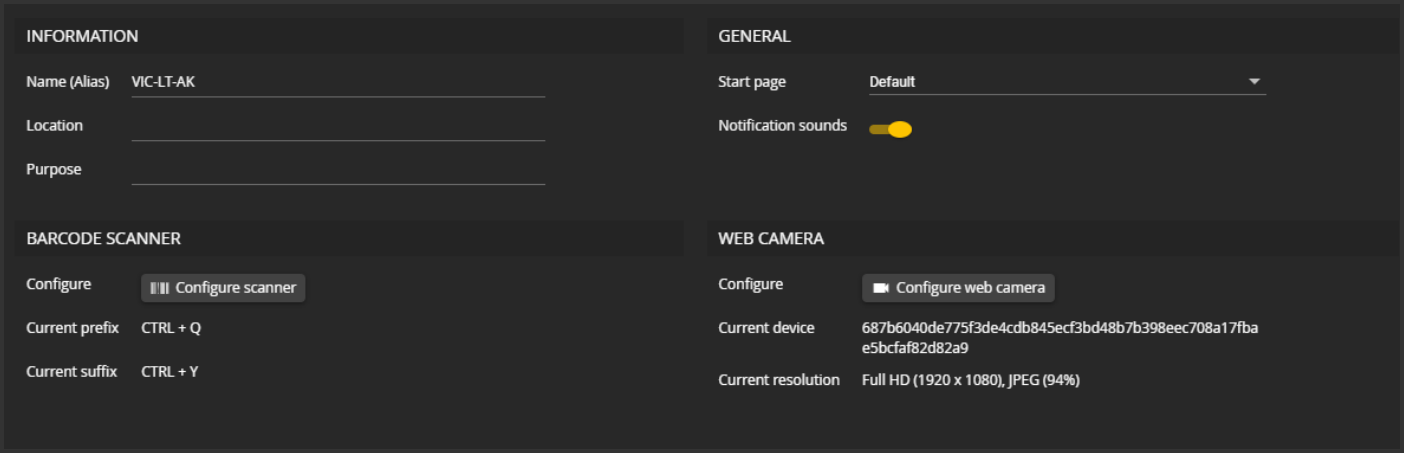
As an alternative to using the WATS layout component for the main layout of a module, the WATS layout header and WATS simple layout components can be used instead. This is the preferred usage in the Reporting WATS module. For other WATS applications, the WATS layout is generally to be preferred.

There are 2 main differences between the two options:

- WLA is only supported for WATS layout.
- When using WATS layout, the content is placed within the component (ng-content), meaning that the header will always stay in a fixed position on the top of the page. When using WATS header & WATS simple layout, the header can also be scrolled if needed.

WATS SECTION COMPONENT (WS)

The WATS section component represents a single block of content on the page. Sections are typically displayed with a header, where the header title describes the content of the section. From a design perspective, pay attention to the padding input of the component, which will provide the appropriate spacing between sections.

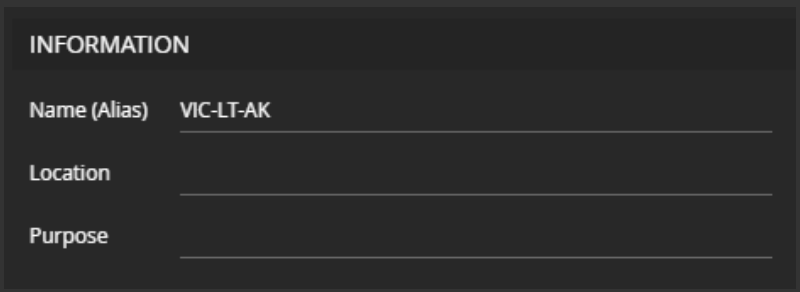


Four WATS sections used to display different areas of Operator interface settings

WATS SECTION ELEMENT COMPONENT (WSE)

The WATS section element component provides a default way of displaying a simple label + content structure, where content typically can be either text or an action (button, select etc.). By default, the section element component adheres to the 32px row height rule, but can be allowed to grow in height if needed.

Note that when placed in a WATS section component, the WATS section elements within it will automatically calculate the appropriate size of the labels in order to align the content. It is also possible to align items across sections, which is encouraged if using multiple sections below each other.



WATS section elements used within a WATS section. Label size is calculated to align input fields.

WATS EDITOR (WE)

The WATS editor component provides a default way of editing an entity. It includes default load / save indicator. The WATS editor should have its own WATS layout, using the default save and cancel buttons. When opening an editor, it will display on top of other content.

WATS ICON (WI)

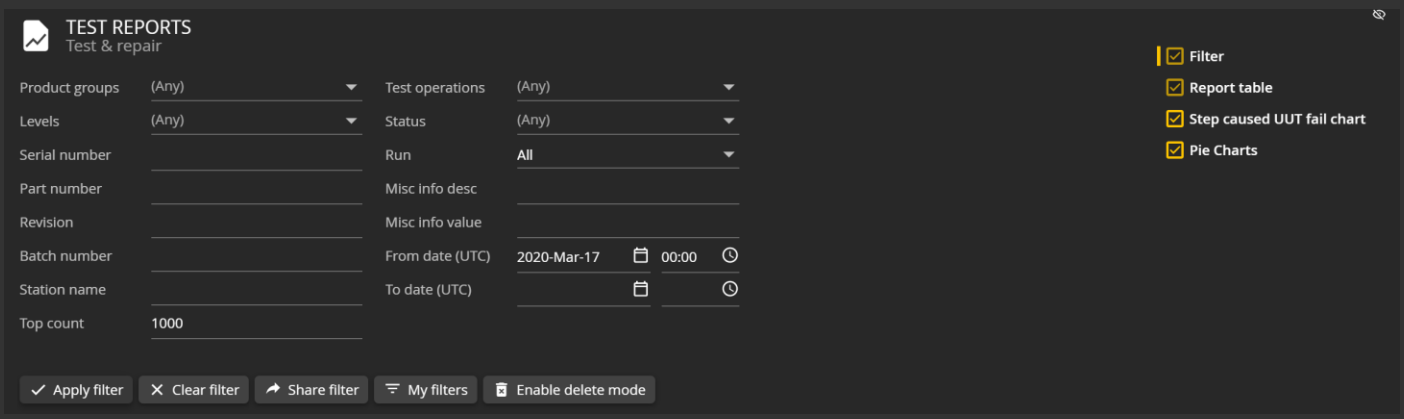
The WATS icon component provides a default way of displaying icons. The main purpose of the WATS icon is less HTML code (than writing the HTML img-tag) and support for themes (show dark icon on light theme and vice versa automatically).

WATS SPLITTER

The WATS splitter is used to split the screen into multiple resizable parts.

WATS NAVIGATOR (WN)

The WATS navigator component provides a quick way to navigate a page and to show/hide different features on a page. The WATS navigator should be utilized for all modules in the Reporting WATS application except for the Dashboard module.



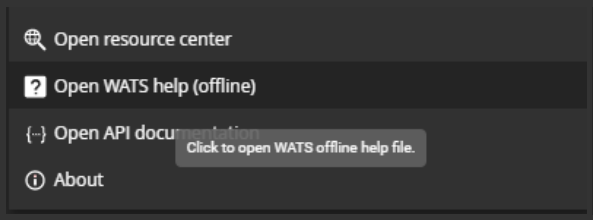
The WATS navigator displayed in the Test reports module, enabling quick navigation and ability to turn on/off charts

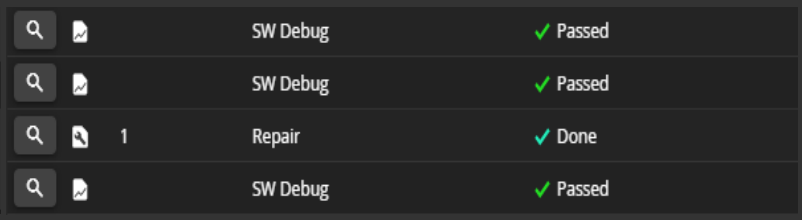
DEFAULT SIZES

The following section describes guidelines for standardized element sizes.

THE 32 PX ROW HEIGHT

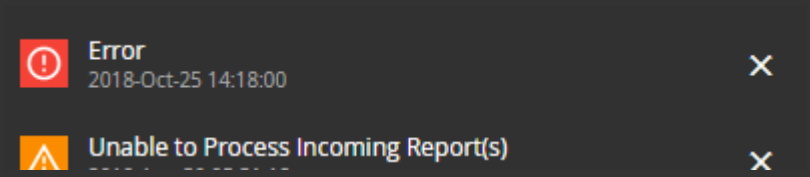
To make the user interface as consistent as possible, applying the same height where appropriate is encouraged. The most common default is a height of 32 pixels. This height is applied to several elements in the WATS Layout and global styling. Examples are: Grid rows, section titles, WATS section elements and drop-down elements. If constructing an element resembling these items, use 32px height if possible.





WATS help menu dropdown, with 32 px item height Grid with row height of 32 px

In some cases, the 32 px row height rule is not feasible. If more vertical space is needed, prefer increasing the size in increments of 16 px, until required space is reached.



Notification drop down item in top bar using a row height of 48px

THE 24 PX ROW HEIGHT (COMPACT)

No row in grid, tables or other tabular elements should have less than 24px height. In general, favor the 32px rule for such items, but for tabular data, a more compact layout will better utilize the available space. For the regular HTML table element, the CSS class .compact can be applied to use 24px row height instead of 32px. When using a table for chart tooltips, favor the .compact rule.

Test statistics	Tested units	Repaired units
Today:	0	0
Last 7 days:	3	1
Current month:	13	5
Current year:	26	9
Total:	2 698	9

HTML table with the .compact class, resulting in 24px row height instead of 32px.

THE 24 PX INTERACTABLE MINIMUM SIZE

No interactable element (buttons, input fields etc.), except for links, should have less than 24px width or height, to ensure sufficient size for touch devices. For raised buttons, a minimum of 32px width is recommended.

ICON SIZES

The following table describes the standard sizes which should be used for icons:

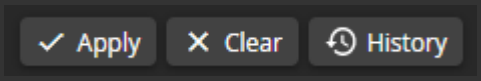
PIXEL SIZE	DESCRIPTION	EXAMPLE USAGE
16x16	Normal icons	Button icons, icons used in conjuction with a line of text
24x24	Emphasised icons	Menu and top bar
32x32	Heading icons	WATS Layout header

DEFAULT SPACING

The following section describes guidelines for standardized space between elements. The spacing will typically be applied by CSS margin or padding. Default CSS classes are available in the global CSS.

5 PX SPACING

Use a spacing of 5 pixels for items which are closely related to each other. Typical examples are space between buttons in the same group of buttons, or between the icon and label within a button.



20 PX SPACING

Use a spacing of 20 px between sections of content. The 20px rule is automatically applied when using the padding input of the WATS section component.

DEFAULT INCREMENTS

As a main rule, use margin and padding in increments of 5. The global CSS provides default classes for padding and margin for 5, 10, 15 and 20 increments, i.e. .padding-right-5, .margin-20 etc.). Except in cases where more than 20px margin or padding is required, favor using these classes.

Z-INDEXES

When creating absolutly or staticly positioned elements utilizing the CSS z-index property, it is important to choose a value which renderes the element on a correct layer in relation to other standard components of the layout. The following table desribes the z-index of common components of the WATS application:

Z-INDEX	COMPONENT
50	WATS editor
100	WATS splitter ghost
200	WATS menu bar
250	Igx overlay
300	Materail overlay
400	WATS preloader
500	Mat tooltip

TYPOGRAPHY
AND LANGUAGES

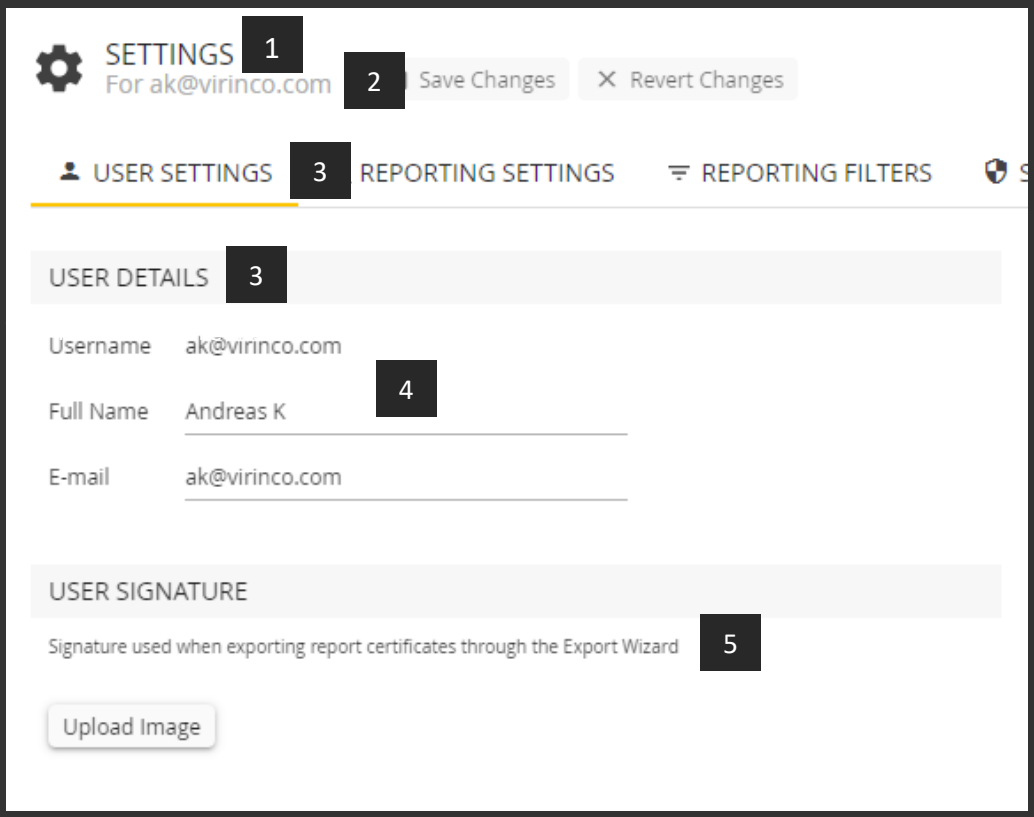
FONT

Use the Google Open Sans font for all designs in WATS

FONT SIZE AND CASE

Favor the following font sizes and transforms when designing components. The described cases referes to the applied CSS text-transform property:

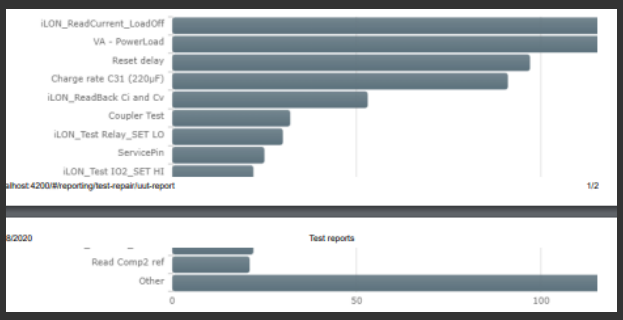
#	TYPE	SIZE	CASE	EXAMPLE USAGE
1	Page header	16 px	Upper	WATS Layout main title
2	Page sub header	14 px	None	WATS Layout sub title
3	Section header	14 px	Upper	WATS Section title or WATS Tab header
4	Body text	12 px	None	Text content of WATS Section or grid
5	Small text	10 px	None	In grids and extra information in a section or section element



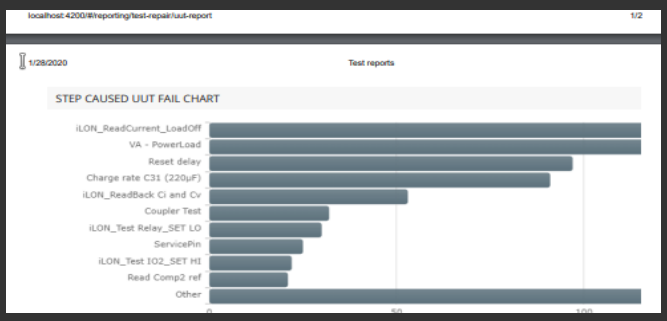
PRINTING

The following describes general guidelines for printing WATS to paper or PDF files with the browser native print dialog.

- By default, all text sizes should be scaled down to 10px for printing. For most elements, this will automatically be applied through the global print CSS.
- For igx grids, add the no-print CSS class the the igx-grid component, and add the grid-print component. Since most igx-grids are virtualized, using the grid-print component, will ensure that the whole grid is printed.
- The total width of the page should be scaled to print on an A4 page.
- Make sure to test each page for its printing capabilities:
    - Apply the no-print CSS class to all elements which should not be shown when printing
    - Apply the avoid-print-page-break CSS class to all elements which should not be broken up over multiple pages.



Igx chart without avoid-print-page-break



Same chart with the avoid-print-page-break CSS class

BOLD TEXT

When displaying a label and value pair, favor using a bold label, and normal font weight for the value. When scanning a page for information, bold text is easier to identify. In the example below, the user is more likely to be scanning the page for the text WATS client version, than the number 6.0.0.119.

Domain	WORKGROUP
Physical memory	>2,025.33 MB
CLR version	4.0.30319.42000
WATS client version	6.0.0.119

Correct usage of bold label and normal weight text.

Another reason for this choice, is that the values often require more space than the labels . Using normal instead of bold weight, will lessen the horizontal space required by the value.

When using a WATS section element, the label will automatically be bold.

Do not use colon at the end of the label. Since the label is bold, and the value is normal weight, this provides the needed separation between the two. Adding colon at the end of the label, simply adds noise to the GUI.

Domain:	WORKGROUP
Physical memory:	>2,025.33 MB

Avoid using colon for labels.

In addition to the guidelines above, also use bold font weight for the following text:

- Grid labels
- General highlighting of a part of the text

0 reports recieved in the last 24 hours
Has been offline for **5 d 21 h 36 m**
Has been inactive for **6 d 21 h 15 m**

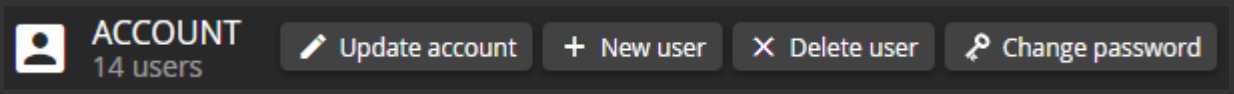
Some parts of a paragraph of text highlighted.

BUTTONS, LINKS AND ACTIONS

The following chapter describes the different clickable actions in WATS, and when to use each. An action in this scenario is a generic term for user interaction with the WATS application user interface:

WATS LAYOUT ACTION (WLA)

Use WATS Layout actions (WLA) when the action pertains to the whole page. The advantage of the WLAs are that they are equally positioned for all modules (see WATS Layout component for more information). A WLA button should be a raised button (see below), unless using a ribbon.



WLA's for the Account module

When determining which actions should be a WLA, consider whether the action is “global” for the page. If an action is more relevant for a part of the page (for instance within a WATS section, keep the button within that area of the page.

RAISED BUTTONS

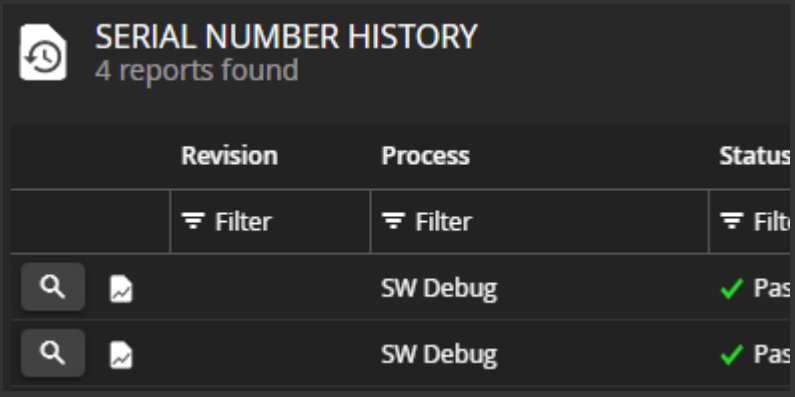
Raised buttons are the general-purpose button in WATS. Use a raised button unless the action falls under the Flat buttons or Link chapters described below, or for the WATS Layout ribbon. The content of each raised button should consist of an icon. In most cases, the icon should be accompanied by a label (see Icon buttons below). Note that spacing between the icon and label is automatically inserted by the global CSS.

A pass all raised button in the Manual inspection module

ICON BUTTONS

If a raised button is placed in a space constricted area, using an icon button is appropriate. The icon button displays an icon, but no label. The most common use case for the icon button is when a button is placed within a grid. When constructing an icon button, follow these guidelines:

- Apply the icon-button CSS class to the button.
- Do not use the big CSS class on an icon button (see the Highlighting, warning and big buttons chapter below for more information).
- Always apply a tooltip to an icon button, as a substitution for the missing label.
- When used in a grid, favor placing the buttons on the left-hand side (first column) of the grid.

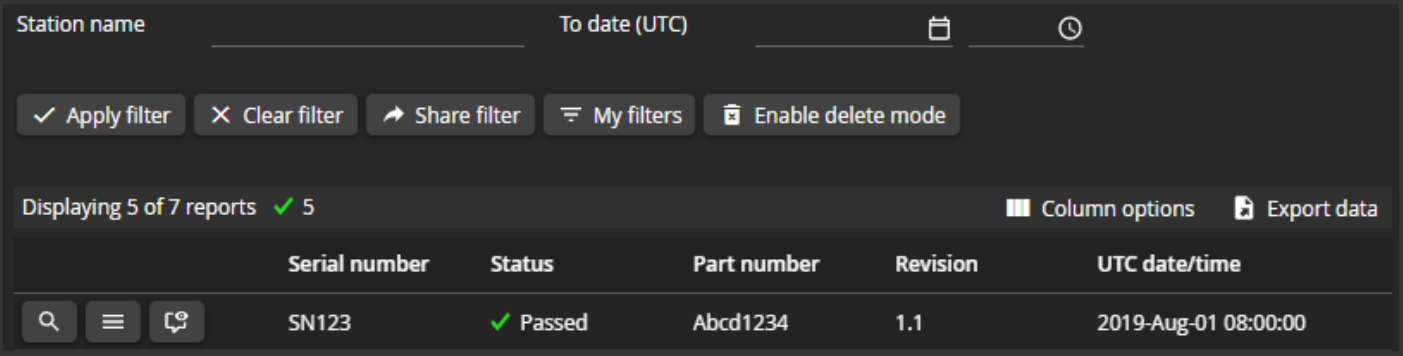


Icon button used in the Serial number history grid

FLAT BUTTONS

Flat buttons are buttons without background, border and shadow. Missing these visual aspects makes them less eye-catching and are therefore suitable for less prominent actions. Use the following guidelines to identify flat buttons on a page:

- Consider whether some actions are less used than others, to identify good candidates for flat buttons.
- Use an icon and label for flat buttons, to make it clearer that the area is a button and not simply text.
- Do not mix raised and flat buttons in the same group of buttons.
- Do not use flat buttons for highlighted, danger or big buttons (see the Highlighting, warning and big buttons chapter below for more information) as this contradicts the idea of a flat button being less prominent.



Correct usage of different button types: more commonly used buttons for filter (raised buttons) and reports (icon buttons) shown alongside less prominent grid column options / export (flat buttons).

HIGHLIGHTED, WARNING AND BIG BUTTONS

Much in the same way that flat buttons make an action less prominent, highlighting and warning styles can be applied to buttons in order to make them more prominent. Highlighting can be applied to any raised button (including WLAs and Icon buttons), but not flat buttons. It is acceptable to use highlighting or warning in conjunction with big buttons. The following table describes the usage:

TYPE	CSS class	EXAMPLE	USAGE
Highlighting	.green	Submit report	Highlighting an action as important..
Warning	.red	Delete	Warning the user that clicking the button will result in dataloss.

Big	.big	▶ Start Test	Extra emphasis on the highlighting or danger. Typically used for a single button, to make it the main action of a page.

LINKS

Use a link (anchor HTML tag) only when applying an action within a paragraph of text. Otherwise apply one of the action types described above.

COMMON ICONS

The following table describes common actions and functionality, and which icons should be used to indicate these. For icon usage not described in this list, try to ascertain whether a similar action or functionality already exists in WATS and use the same icon as previously used if found, in order to promote conformity:

USAGE	WATS ICON NAME	ICON
Add	add	
Collapse	collapse	
Confirm / accept	check	
Copy / clone	copy	
Date	calendar	
Draggable	drag	
Close / cancel / clear *	clear	
Delete *	clear-red	
Download	download	
Edit	edit	
Expand	expand	
Link to page outside WATS	external	
Filter	filter	
Search / view details	search	
State: error **	error or error-red	
State: ok **	check-circle or check-circle-green	
State: warning **	warning or warning-orange	
Time	clock	
Upload	upload	
View help	help	
View information	information	

* Use the clear-red icon for actions which will result in data loss, typically deleting data from the server. For other uses, use the clear icon.

** For state icons, favor the colored versions to better indicate the state.

COMPONENTS
USAGE GUIDE

The following table shows which components to choose from when designing a page in WATS. In addition to these components, pay attention to the Layout chapter, which describes the general components to use for designing the layout of a page. Note that this guide is a general overview, and final decisions about components should be made in a design meeting. Components not mentioned in either this or the Layout chapter, should not be used without revising the design document.

COMPONENT	LIBRARY	NOTES
Autocomplete	Material	
Avatar	Igx	
Badge	Igx	
Button	Igx	Both flat and raised buttons (see the buttons chapter for more information on when to use each type)
Chart	Igx	
Checkbox	Igx	
Combo	Igx	
Dialog	Igx	Note: for simple “ok” and “yes/no” dialogs, use the core DialogService to awoid creating a new dialog in template.
Date picker	Igx	
Dropdown	Igx	
Expansion panel	Igx	Use for list of items ect, not for page layout.
Grid	Igx	Including hierarchical grid and tree grid.
Icon	WATS or igx	
Input	Material	
List	Igx	
Spinner	Material	
Radio button	Igx	
Select	Material	
Sidenav	Igx	
Splitter	WATS	

Switch	Igx	
Tabs	Material and WATS	Use Material tabs component, with WATS tab header component for the header template.
Time picker	Igx	
Tooltip	Material or igx	Use matieral for all text based tooltips. Use igx for more advanced tooltips requiring html templating. Use a show delay of 1000ms and a no hide delay as default. For text or other non-interactive content in a grid cell, use 500ms.
Tree	Material	

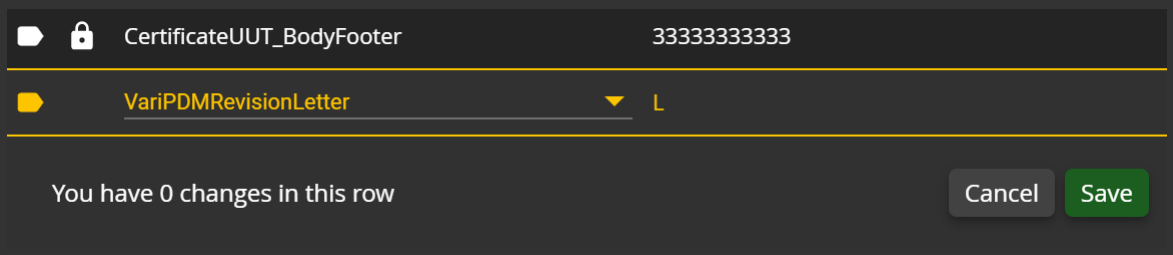
DATA EDITING

When designing modules which allows data input and/or editing, follow the guidelines below for determining the structure of the module.

When performing edit of one or more entities, always provide the user with options to either save or cancel the changes made.

GRID EDITING

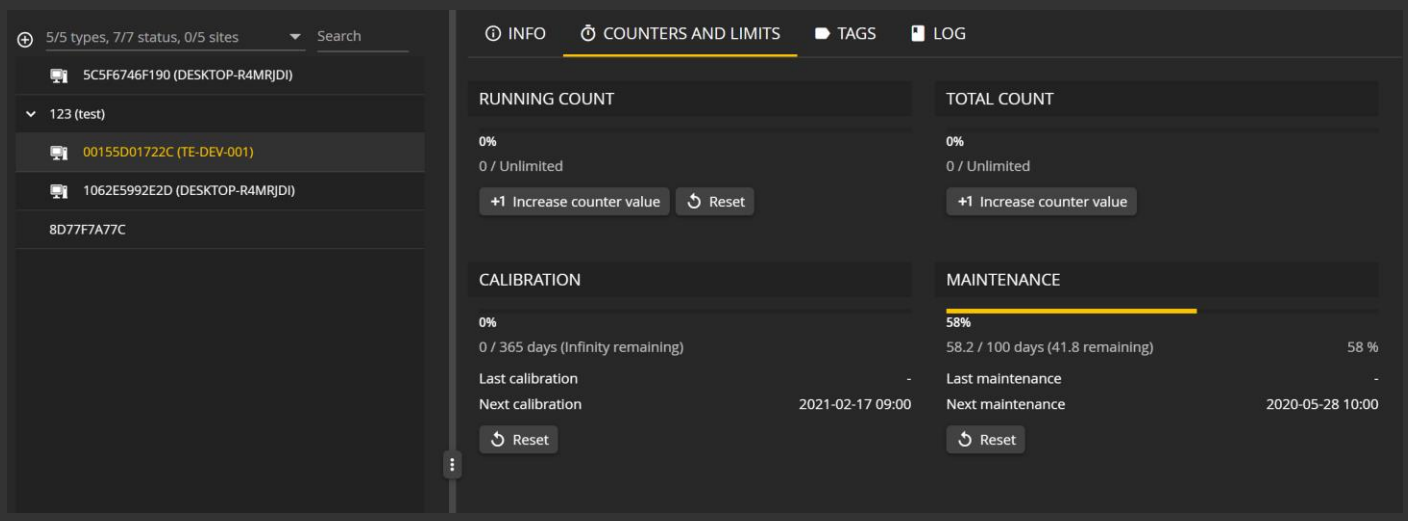
Grid editing refers to editing done to a row within a grid. Favor grid editing when all the editable data already is present in the grid. Typically, this is suitable for smaller amounts of data. For grid editing, use the build in editor of the igx-grid.



Editing a row in a grid in the Product manager module

SPLIT EDITING

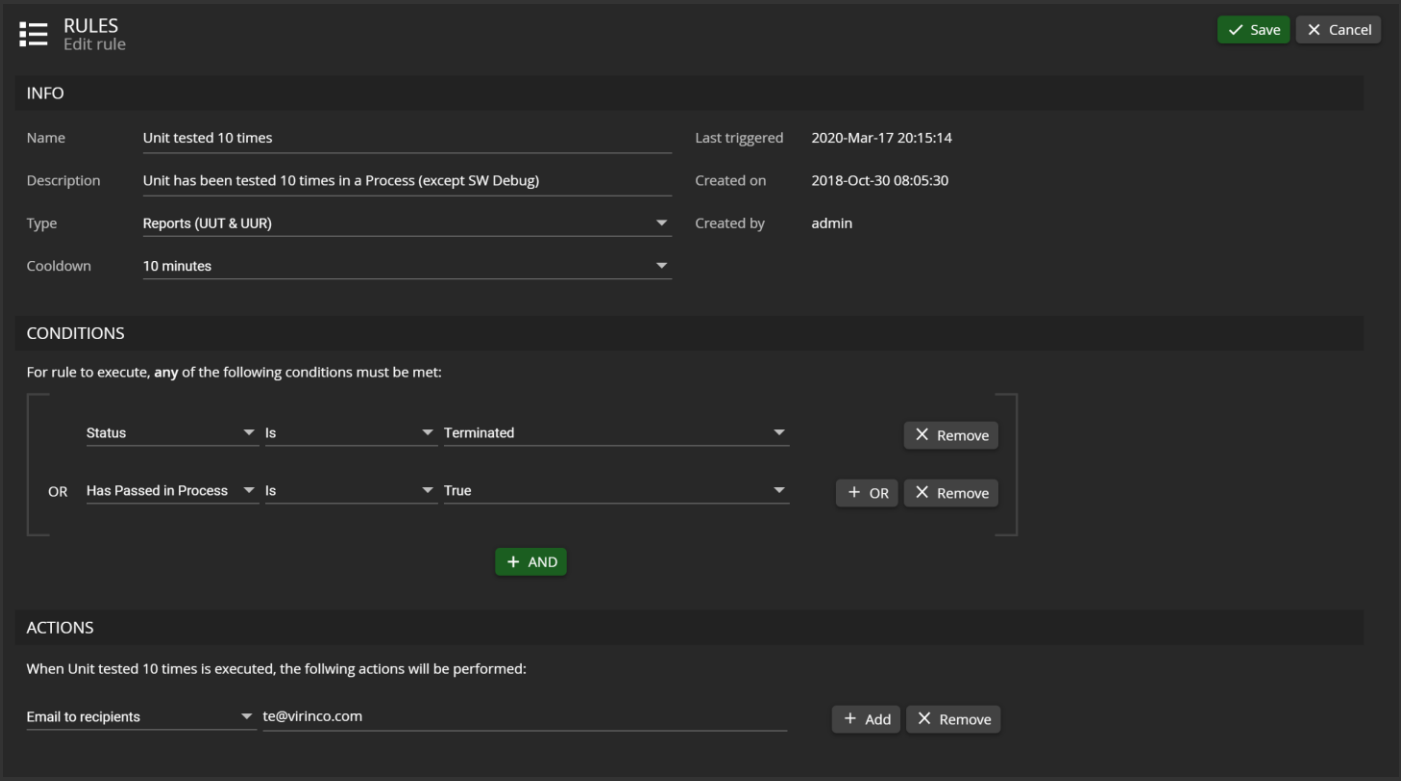
A split editing setup is done by dividing the screen into a left and a right part. The left part typically consists of a grid, list or tree component, from which an entity can be selected. The right part of the screen handles editing of the selected entity. The two parts should be divided by a WAST splitter.



Split editing for the Asset manager

WATS EDITOR

If neither of the above editing options are suitable, use the WATS editor. The WATS editor is sutiable when the entity or enities being edited contains a lot of properties and/or requires a large percentage of screen real esteate.



Editing rules in the alarms and notifications modules requires a large percentage of the screen and utilizes the WATS editor.

USER FEEDBACK

As a main rule, when triggering operations which performs calls to the server. The user should be notified of the ongoing operation. It is recommended to utilize one or more of the following standard ways of notification:

Top bar indicating a long running API call

As default, the top bar will display a loading / saving indication for all API calls which last more than 1.5 seconds. No additional actions are required when implementing a module, as long as the top bar is present.

Using an additional load / save indicator, typically located at the part of the screen where the user has his/her focus improves usability. As a general rule, a load / save indicator should consist of a spinner and a short text explaining the operation being performed.

In addition to the top bar loader, multiple standard load / save indicators exist in WATS. It is recommended to use one of the default indicators where possible.

The following shared WATS components support standardized load / save indication:

#	TYPE	USAGE
1	WATS loader	General load / save indicaion for the whole or part of a module.
2	WATS editor	Edit entities full screen
3	Spinner button	General load / save indication for buttons

GRID
GUIDELINES

The following guidelines applies when designing a igx-grid component. Where applicable, the guidelines also applies to the igx-hierarcical-grid and igx-tree-grid components.

- Favor using the standard WATS grid toolbar to enable choosing columns (see grid service below) and prodviding export to Excel, CSV and TSV. By default, all grids should supply the option to export to these formats. Column options (choosing columns and toggling grouping) should be enabled for all grids in the Reporting WATS module, but may be omitted in other WATS modules.
- Favor using the standard WATS grid service. This will give the following properties to the grid:
    - Support automatic grid sizing, including support for fixed column sizes.
    - Save column order
    - Save visible columns (in conjuction with the WATS grid toolbar.
- Favor icon buttons when using buttons inside a grid.
- Favor buttons on the left side of the grid.
- When designing grids, do not enable the filtering feature, if the page allready contains a WATS filter component.
- When designing grids, use the default grid loader and grid templates components, to make sure grids look the same when loading and when not containing any data.

GRID COLUMNS

The following table describe the naming conventions, order, and default sizes of common grid columns:

COLUMN NAME	MIN PX SIZE	NOTES
Product group		
Level		
Period		
Serial number		
Type		Report type UUR or UUR
Status		Enough space to show longest status
Part number		
Product name		
Revision		
Batch number		
UTC Start date/time		
Start date/time		
UTC first seen date/time		
UTC last seen date/time		
Test operation		
Test operation code		
Repair operation		
Repair operation code		
Software filename		
Software version		
Operator		
Station name		
Location		
Purpose		
Fixture id		
Test socket index		
Test Socket count		
Error code		

Error message		
Comment		
Misc info desc		
Misc info value		
Report receive count		
Report size (KB)		
Passed in run		
Step caused UUT fail		
Step caused UUT fail path		
Connection time (avg)		
Connection time (min)		
Connection time (max)		
Connection time (total)		
Execution time (avg)		
Execution time (min)		
Execution time (max)		
Execution time (total)		
Running time		
Available time		
Availability		OEE
Performance		OEE
Quality (LPY)		OEE
OEE		OEE
Week count		
Target output		
Unit #		Non-removable, Enough space to always show header (english)
Unit # +/-		
Previous period Unit #		
FPY pass #		Enough space to always show header (english)
FPY fail #		Enough space to always show header (english)
FPY %		Enough space to always show header (english)
FPY +/-		Enough space to always show header (english)

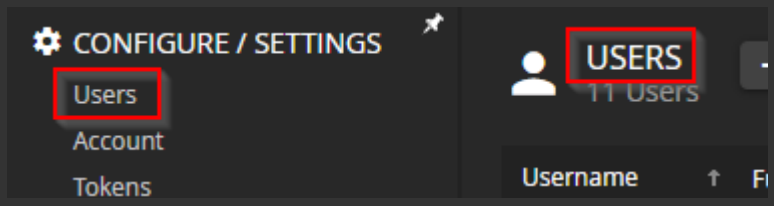
FPY RTY		Enough space to always show header (english)
FPY PPM		Enough space to always show header (english)
FPY PPM +/-		
FPY warning		Enough space to always show header (english)
FPY alarm		Enough space to always show header (english)
SPY pass #		Enough space to always show header (english)
SPY fail #		Enough space to always show header (english)
SPY %		Enough space to always show header (english)
SPY +/-		Enough space to always show header (english)
SPY RTY		Enough space to always show header (english)
SPY PPM		Enough space to always show header (english)
SPY PPM +/-		
SPY warning		Enough space to always show header (english)
SPY alarm		Enough space to always show header (english)
TPY pass count		Enough space to always show header (english)
TPY fail #		Enough space to always show header (english)
TPY %		Enough space to always show header (english)
TPY +/-		Enough space to always show header (english)
TPY RTY		Enough space to always show header (english)
TPY PPM		Enough space to always show header (english)
TPY PPM +/-		
TPY warning		Enough space to always show header (english)
TPY alarm		Enough space to always show header (english)
LPY pass #		Enough space to always show header (english)
LPY fail #		Enough space to always show header (english)
LPY %		Enough space to always show header (english)
LPY +/-		Enough space to always show header (english)
LPY RTY		Enough space to always show header (english)
LPY PPM		Enough space to always show header (english)
LPY PPM +/-		
LPY warning		Enough space to always show header (english)
LPY alarm		Enough space to always show header (english)

Test Report #		Non-removable, Enough space to always show header (english)
Test Report # +/-		
Previous period Test Report #		
TRY pass #		Enough space to always show header (english)
TRY fail #		Enough space to always show header (english)
TRY %		Enough space to always show header (english)
TRY +/-		Enough space to always show header (english)
TRY PPM		Enough space to always show header (english)
TRY PPM +/-		
TRY warning		Enough space to always show header (english)
TRY alarm		Enough space to always show header (english)
Retest #		
Repair Report #		
Repair Report # +/-		
Previous period Repair Report #		

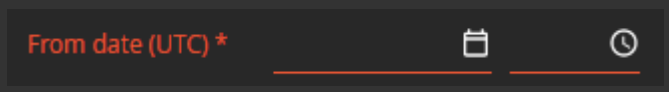
GENERAL GUIDELINES

The following list describes miscellaneous guidelines for designing modules in WATS:

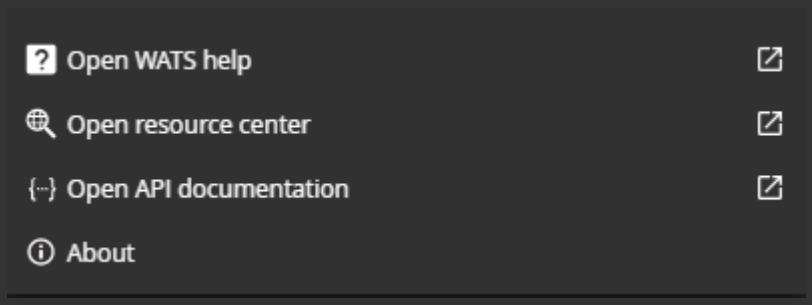
- Avoid using screenshots inside the WATS application, as screenshots add extra maintenace work in order to keep them current.
- When designing icons, always provide both a #fff and #424242 (-dark) versions of the icons.
- When providing a title for a module, make sure that other parts of the application refrences the name / title in a consistent way (menu item, navigator element, page title ect.)



- When an input field is required, mark the label with a star. If the user fails to provide an input value, mark the field with the red-text CSS class. The WATS section element has built in support for this behavior.



- For actions resulting in opening a page outside WATS, favor using external WATS icon on the right-hand side of the action



3 out of 4 drop down menu items leads to external pages.